

Astro-1_up_ALL CC

G protein-coupled receptor complex

postsynaptic membrane

GABA receptor complex

synaptic membrane

GABA receptor complex

synaptic membrane

Astro-1_up_ALL BP

negative regulation of
adenylate cyclase activity

negative regulation of lyase activity

negative regulation of lyase activity

regulation
of adenylate
cyclase
activity

synaptic vesicle docking

synaptic vesicle docking

vesicle docking
involved in exocytosis

neuron-glia cell signaling

neuron-glia cell signaling

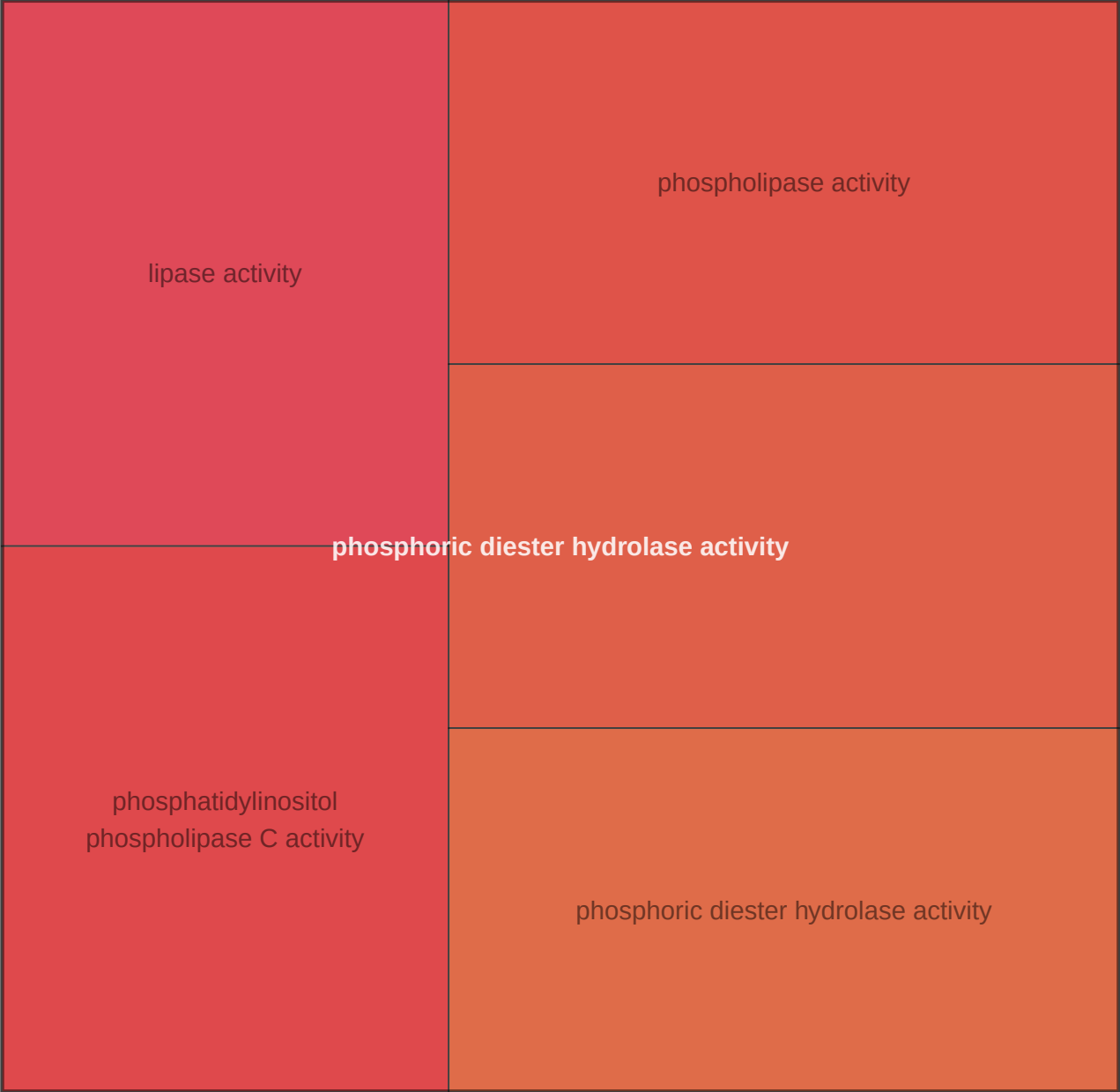
gamma-aminobutyric
acid signaling pathway

lipid import into cell

Astro-1_up_ALL MF



Astro-2_down_ALL MF



Astro-2_up_ALL MF



Astro-3_down_ALL CC

compact myelin

compact myelin

myelin sheath

Astro-3_down_ALL BP

axon ensheathment

ensheathment of neurons

iron ion transport

myelination

iron ion transport

myelination

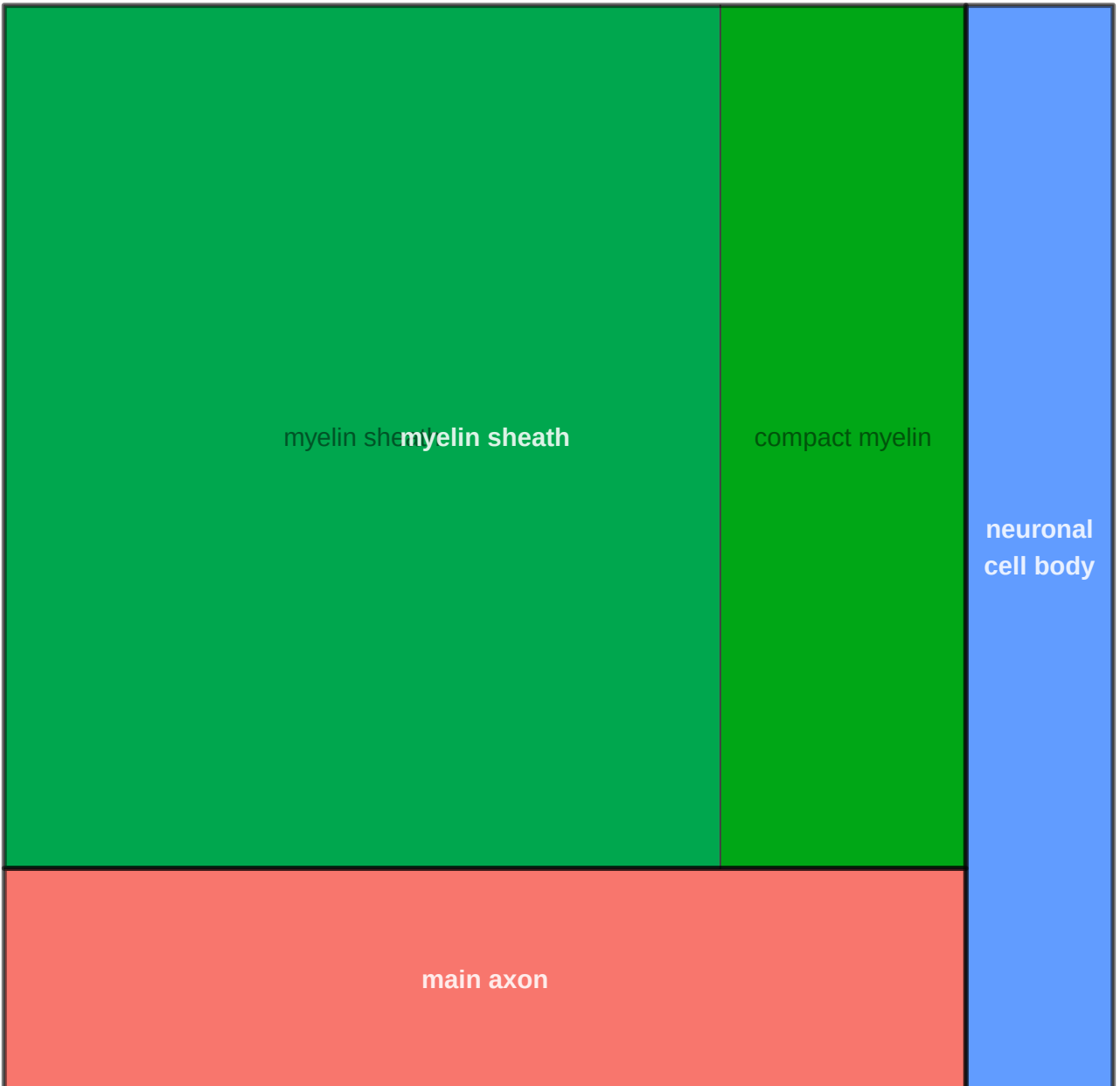
myelin assembly

regulation of
vesicle-mediated
transport

Astro-3_down_ALL MF

active transmembrane transporter activity		alcohol transmembrane transporter activity		amine transmembrane transporter activity		antiporter activity		ferric iron binding	
calcium, potassium:sodium antiporter activity		secondary active transmembrane transporter activity		calcium:sodium antiporter activity		retinoic acid-responsive element binding		structural constituent of myelin sheath	
choline transmembrane transporter activity		calcium:monoatomic cation antiporter activity		bile acid transmembrane transporter activity		immunoglobulin binding		metallochaperone activity	
organic cation transmembrane transporter activity		sodium-independent organic anion transmembrane transporter activity		solute:potassium antiporter activity		sialic acid binding			
CD4 receptor binding		transferrin receptor binding		metallodipeptidase activity		dipeptidase activity			

Astro-3_down_M CC



Astro-3_down_M BP

myelin assembly

ensheathment of neurons

myelin assembly

axon ensheathment

myelination

positive
regulation
of bone
resorption

regulation of endocytosis

regulation of endocytosis

regulation of vesicle-mediated transport

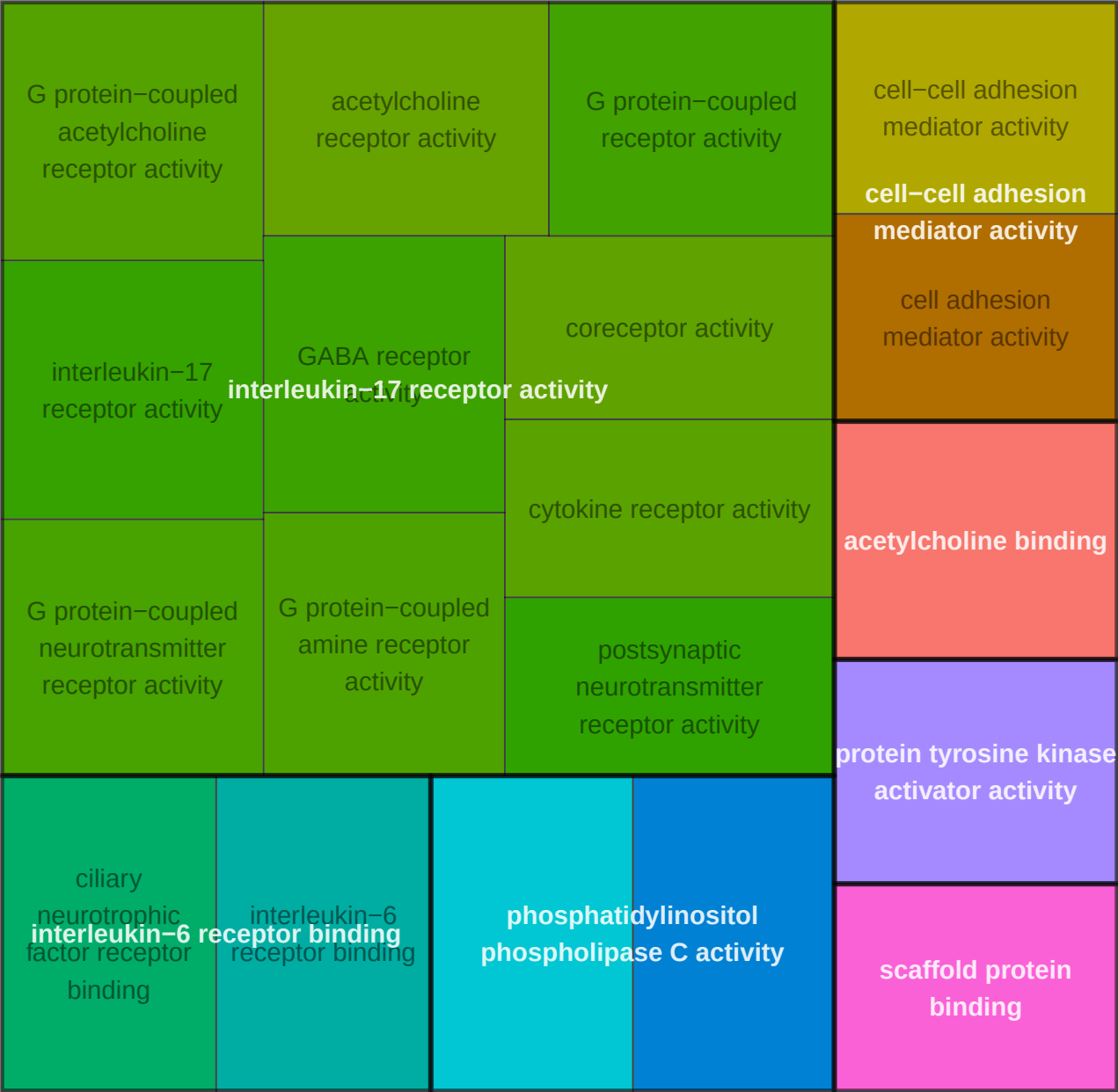
Astro-3_up_ALL CC



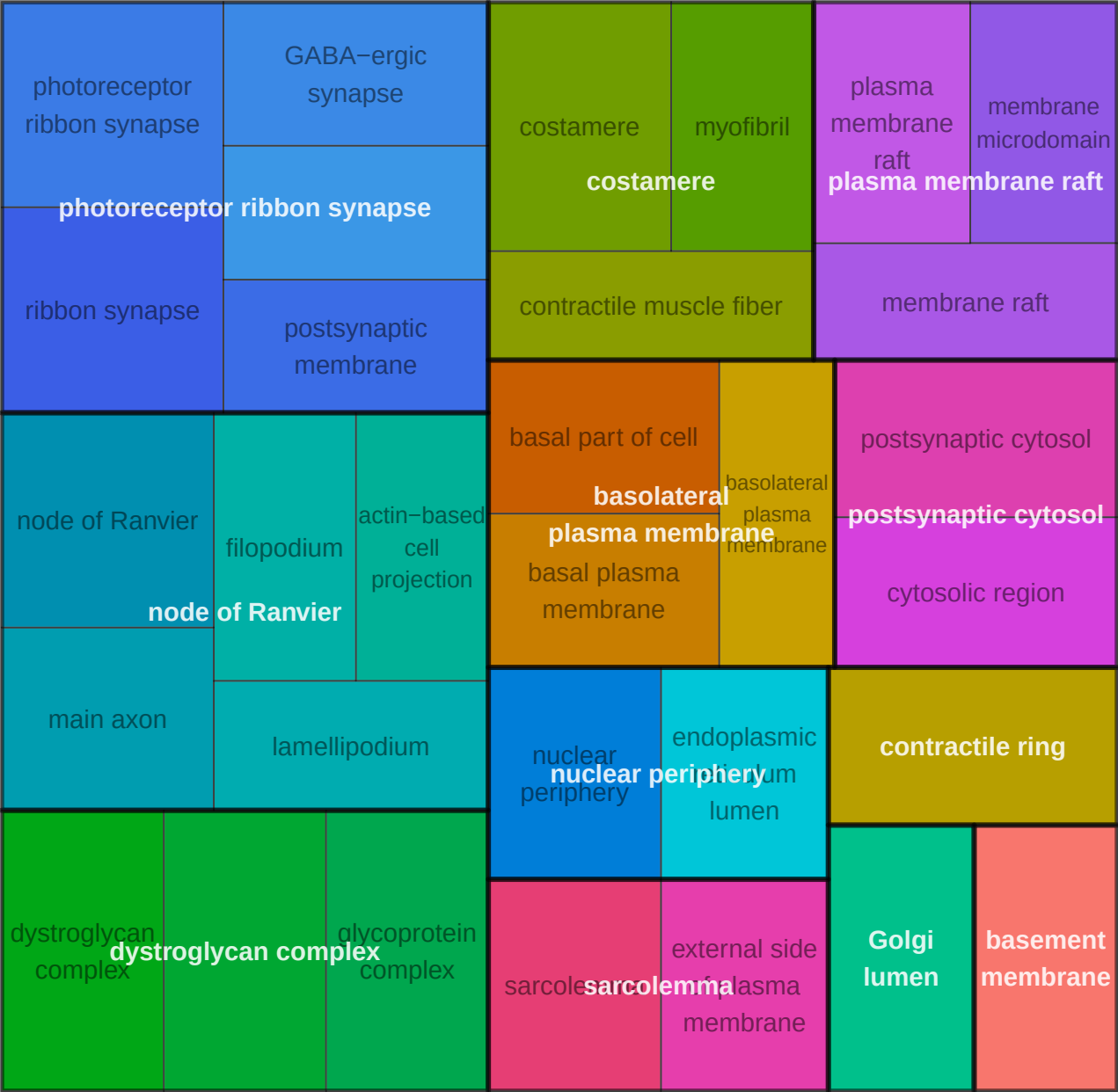
Astro-3_up_ALL BP



Astro-3_up_ALL MF



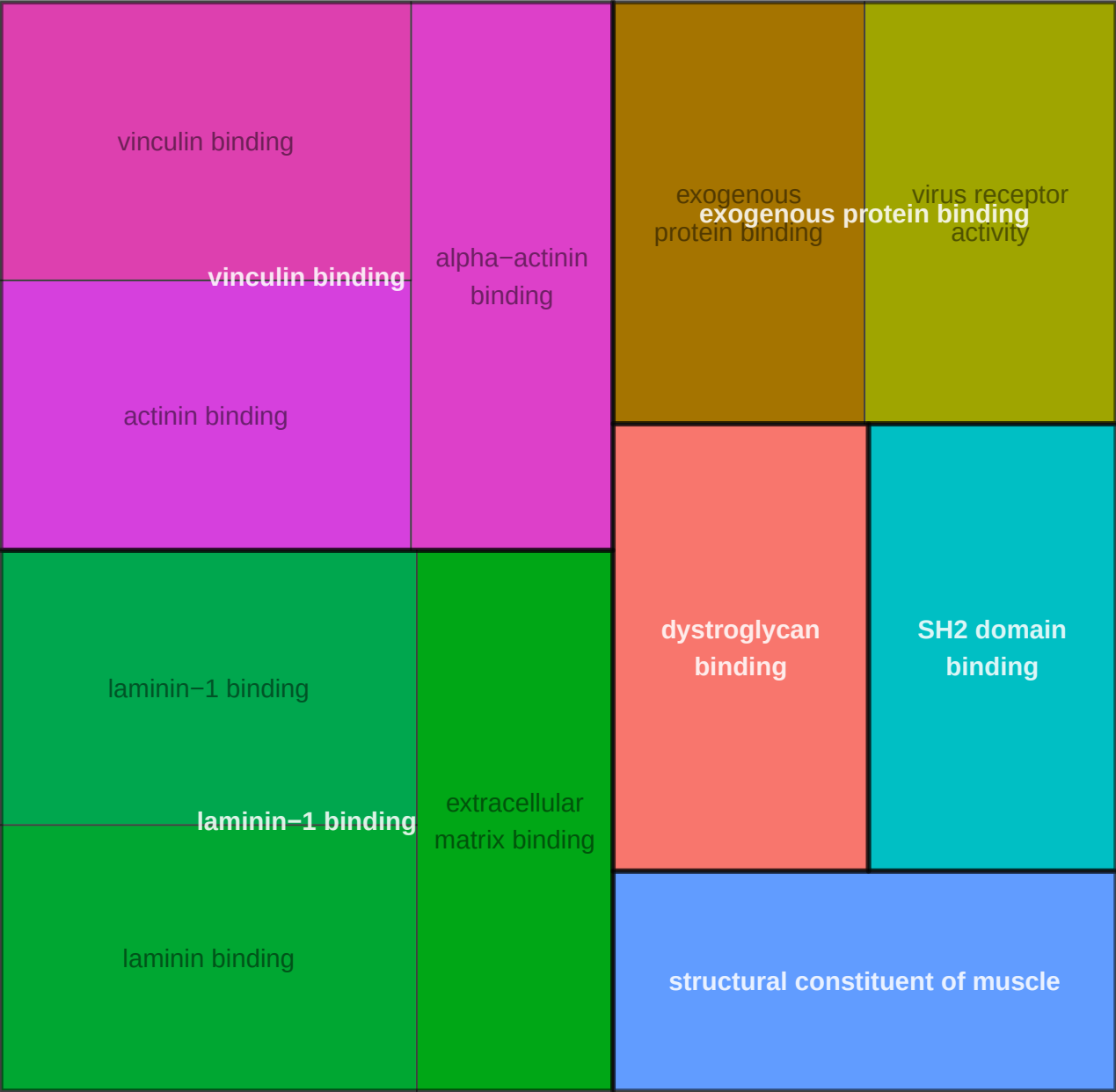
Astro-3_up_M CC



Astro-3_up_M BP



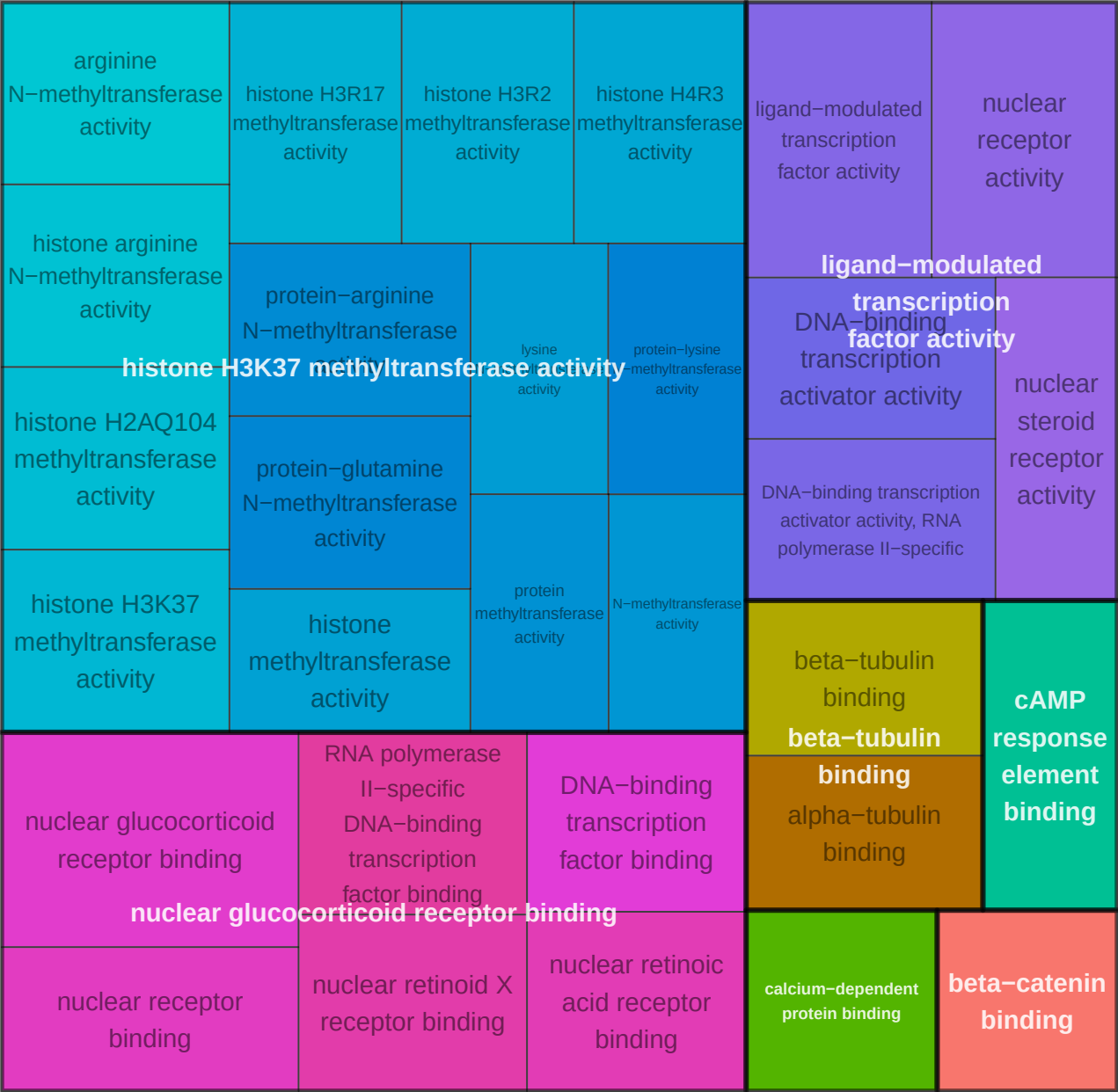
Astro-3_up_M MF



Astro-4_up_F BP



Astro-4_up_F MF



Excit-L2_5-1_down_ALL CC

dendritic spine

neuron spine

neuron spine

Excit-L2_5-1_down_ALL MF



Excit-L2_5-1_up_M CC

specific granule membrane

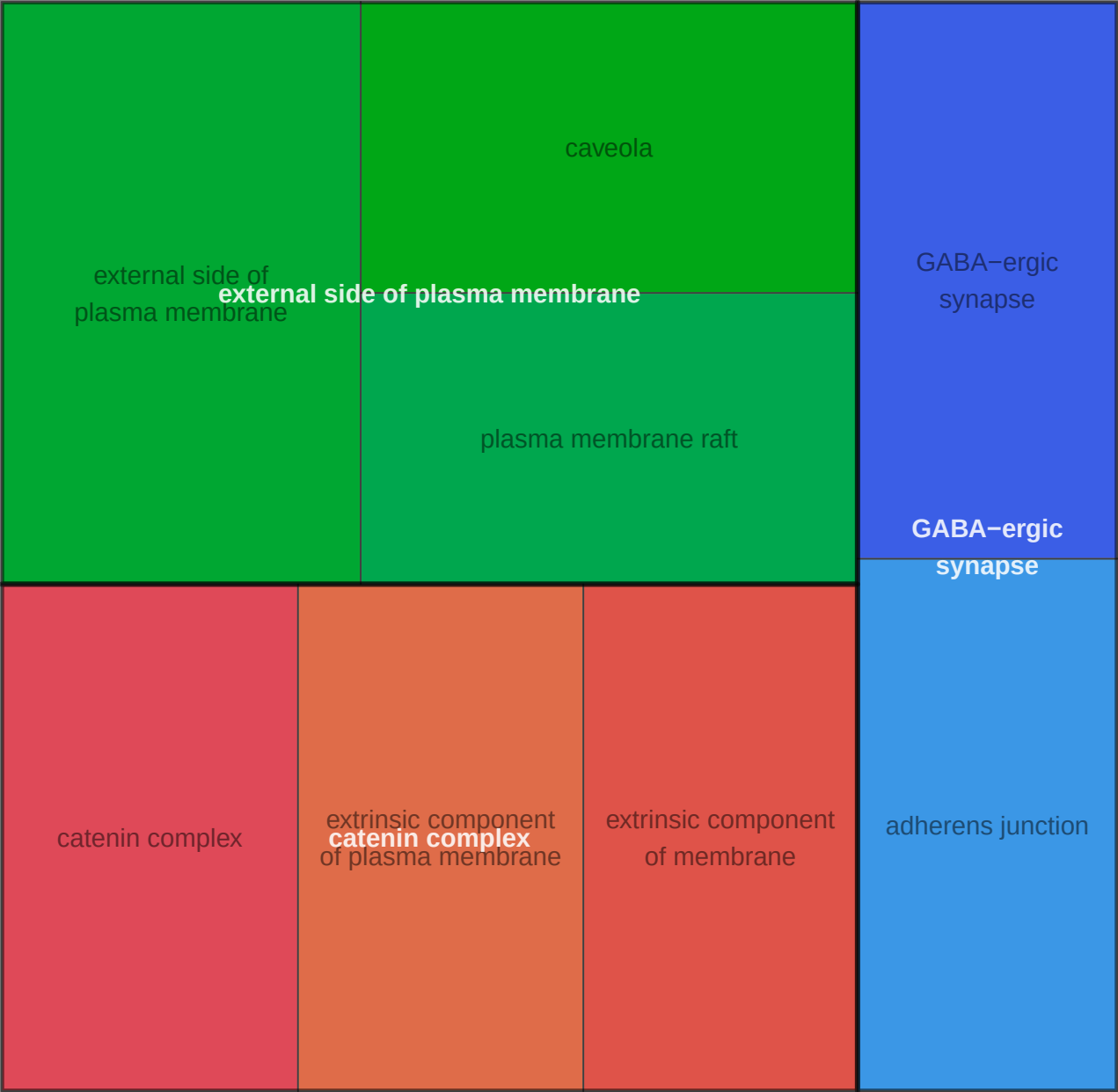
specific granule membrane

secondary granule membrane

Excit-L2_5-1_up_M MF



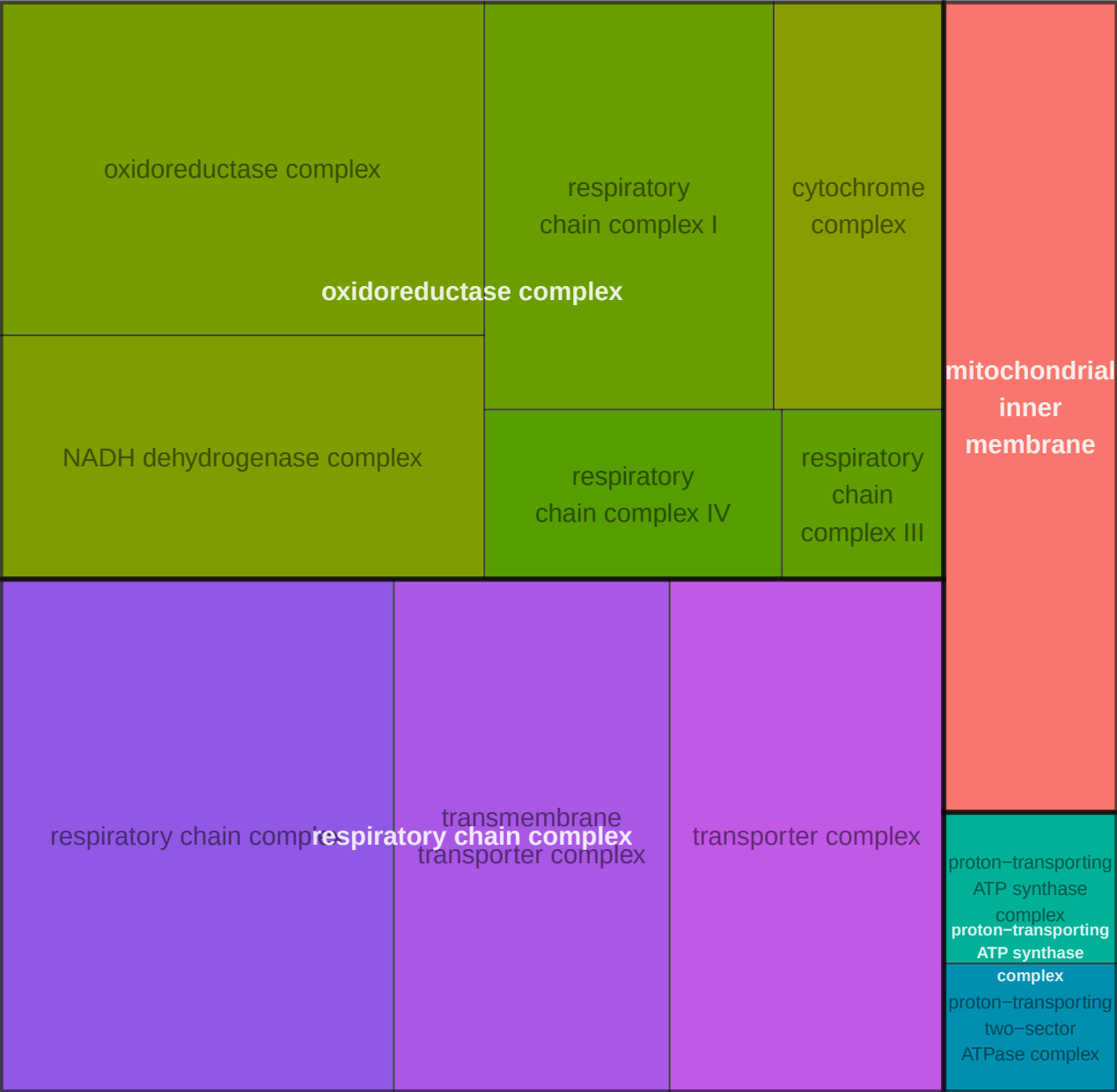
Excit-L2_5-2_down_M CC



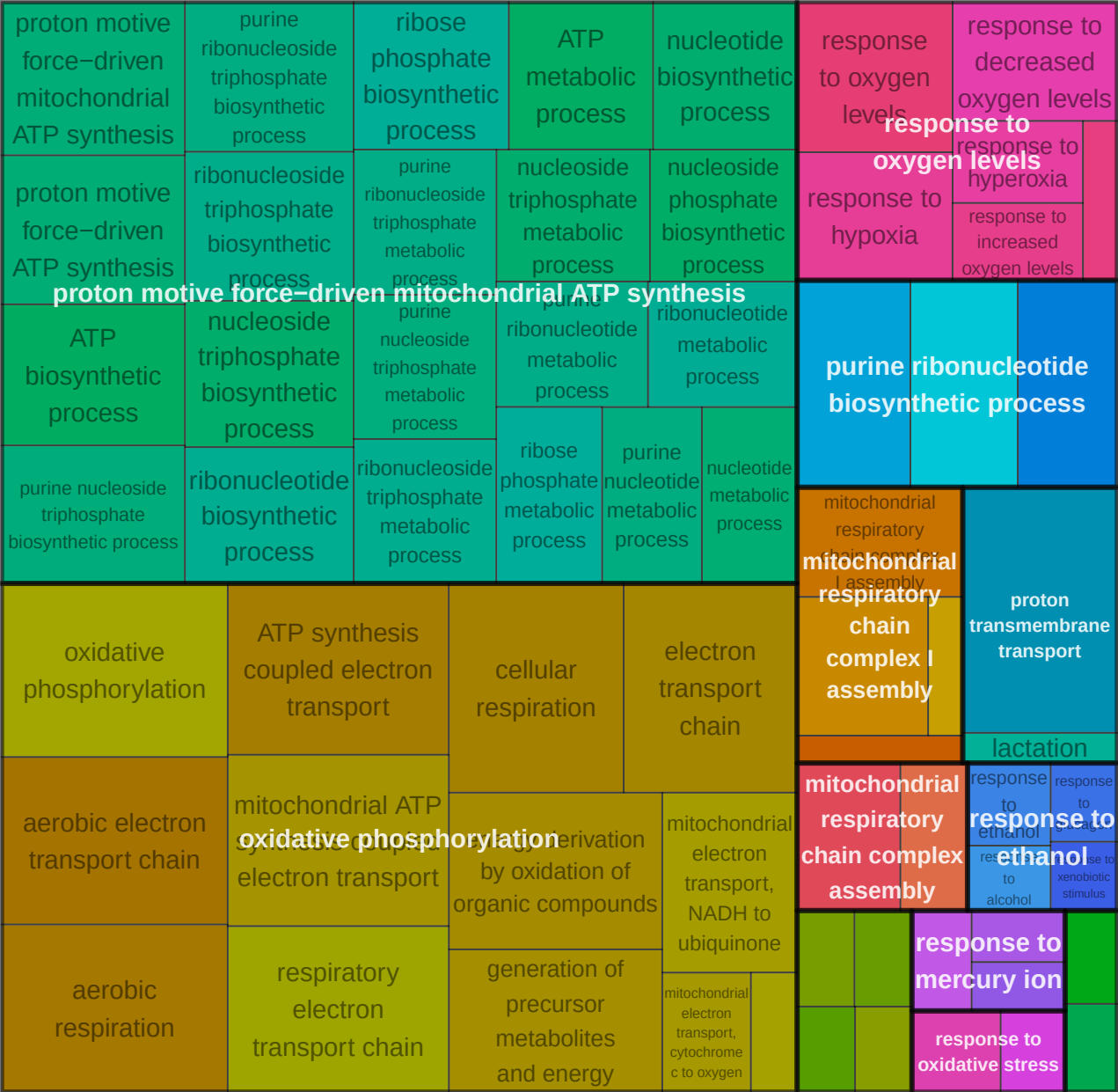
Excit-L2_5-2_down_M BP



Excit-L5-2_down_ALL CC



Excit-L5-2_down_ALL BP



Excit-L5-2_down_ALL MF



Excit-L5-2_down_M CC

germ cell nucleus

germ cell nucleus

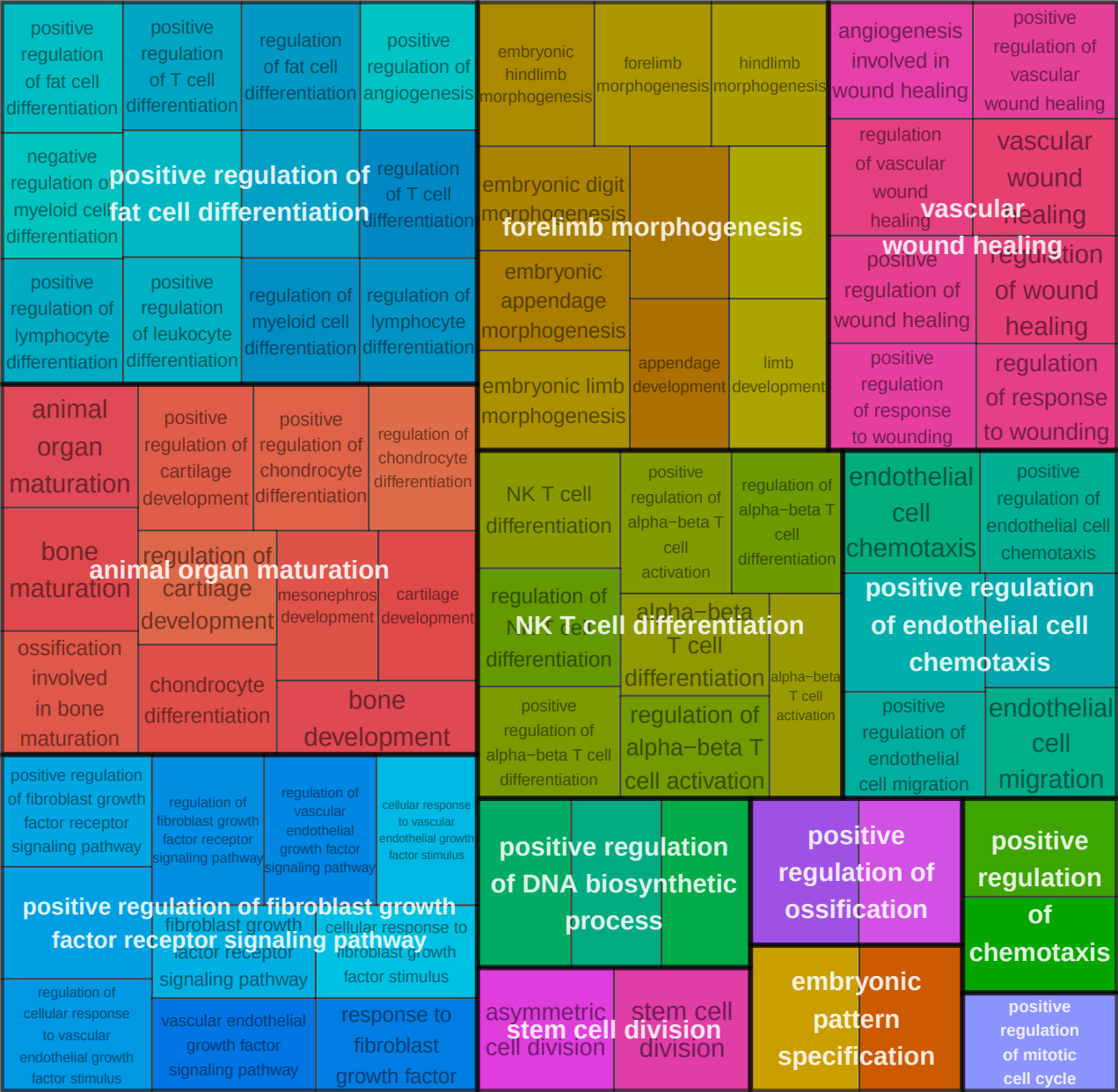
male germ cell nucleus

PML body

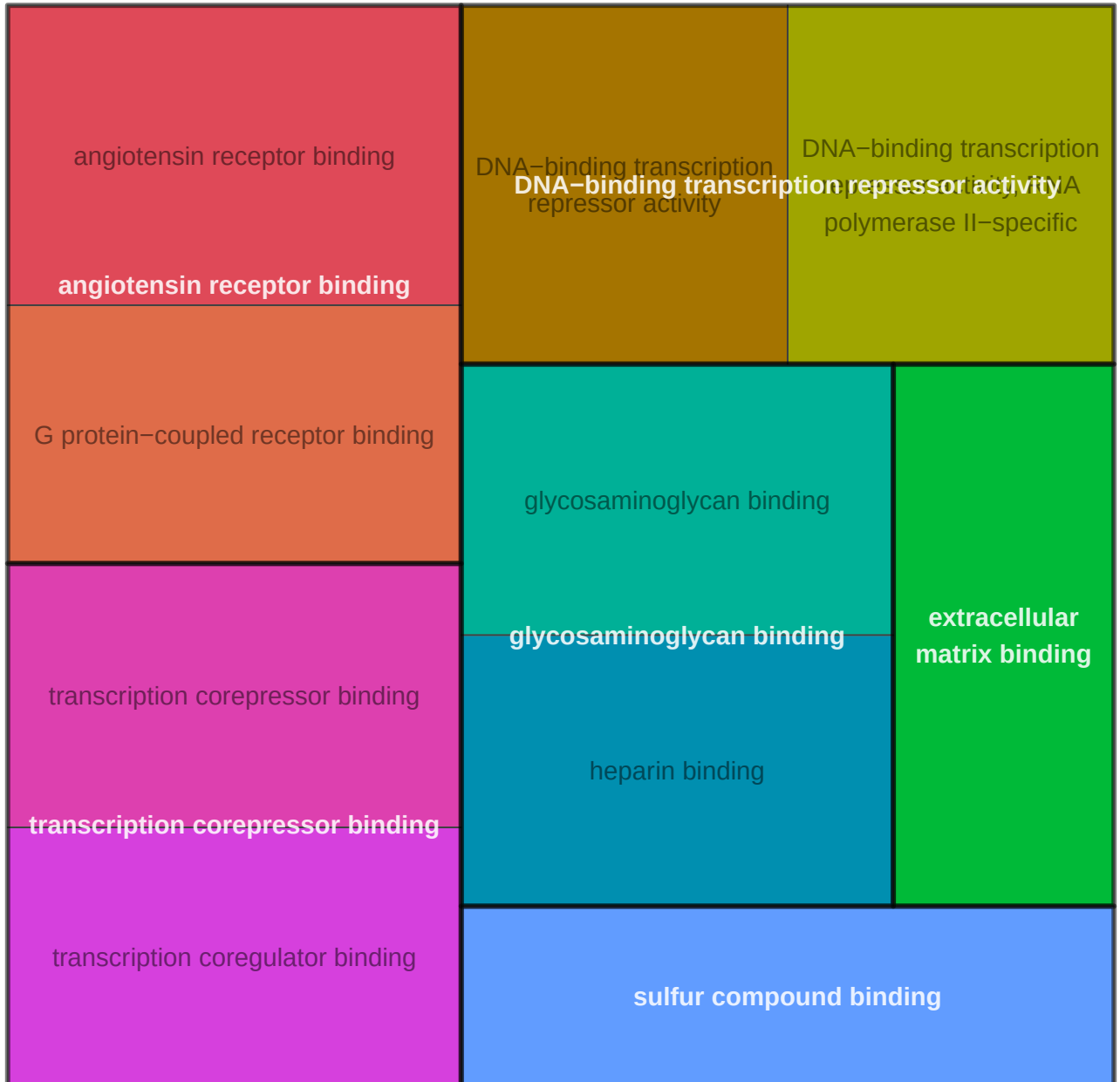
transcription repressor complex

basement membrane

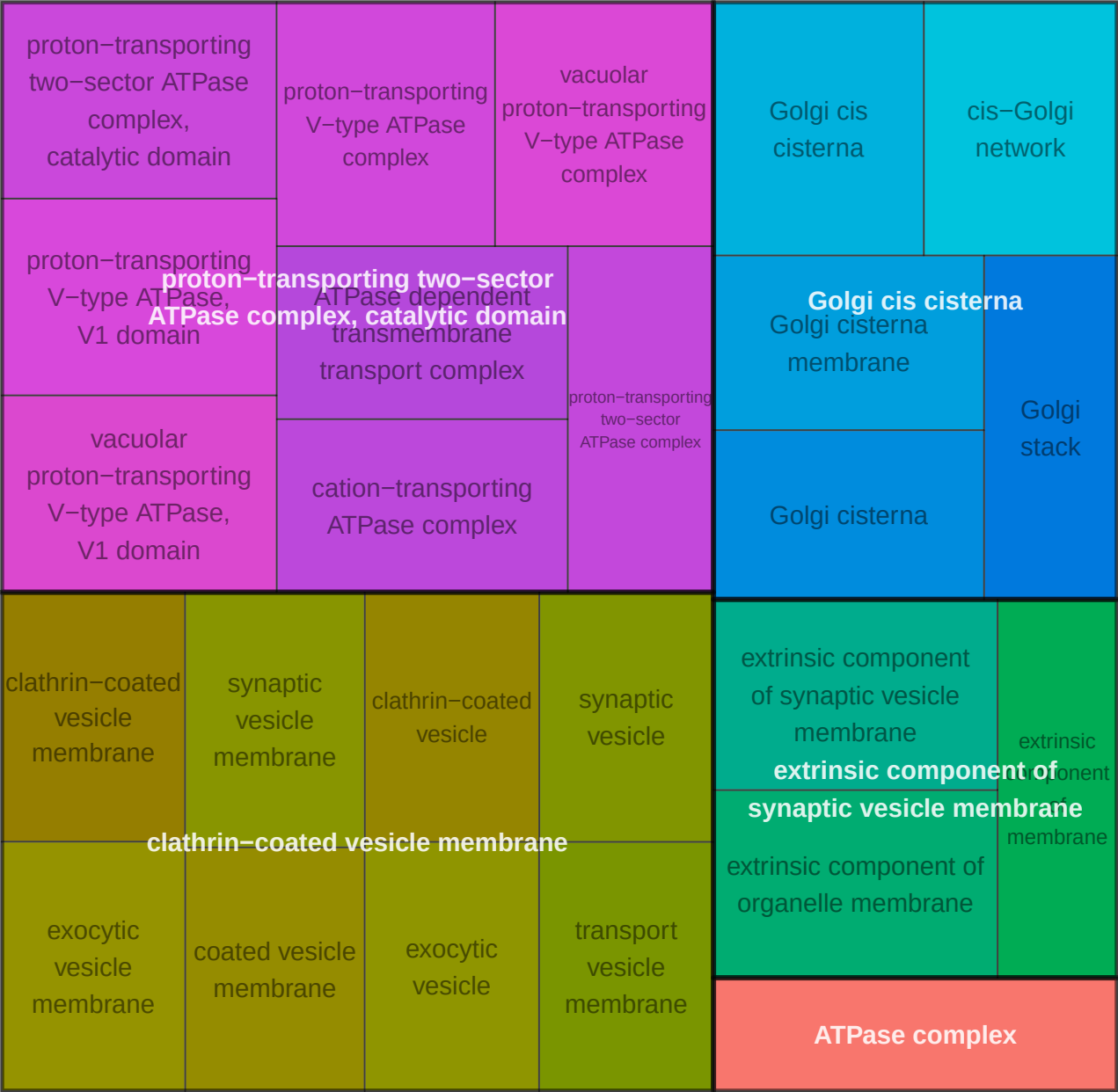
Excit-L5-2_down_M BP



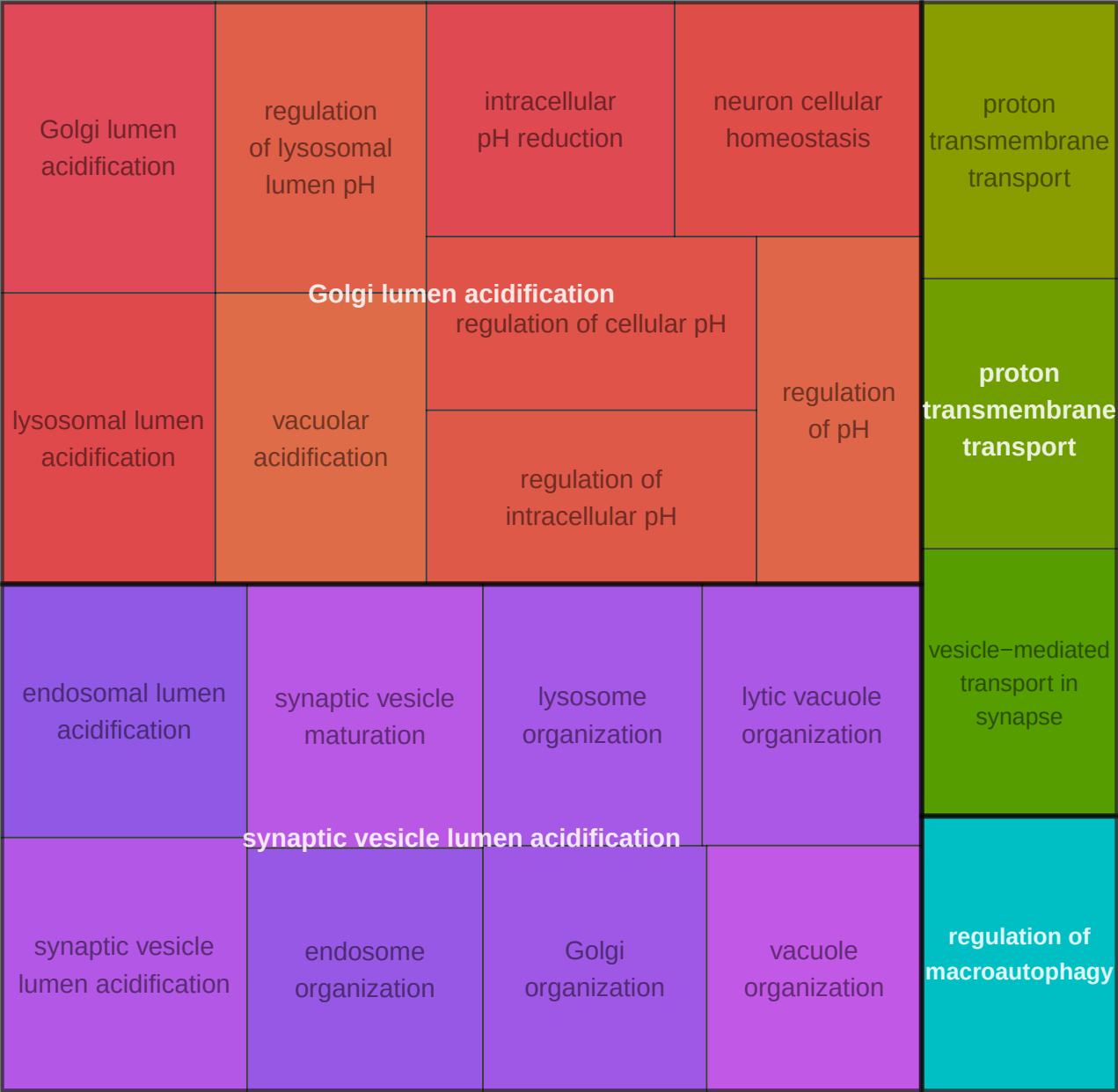
Excit-L5-2_down_M MF



Excit-L5-2_up_ALL CC



Excit-L5-2_up_ALL BP



Excit-L5-2_up_M CC

adherens junction

adherens junction

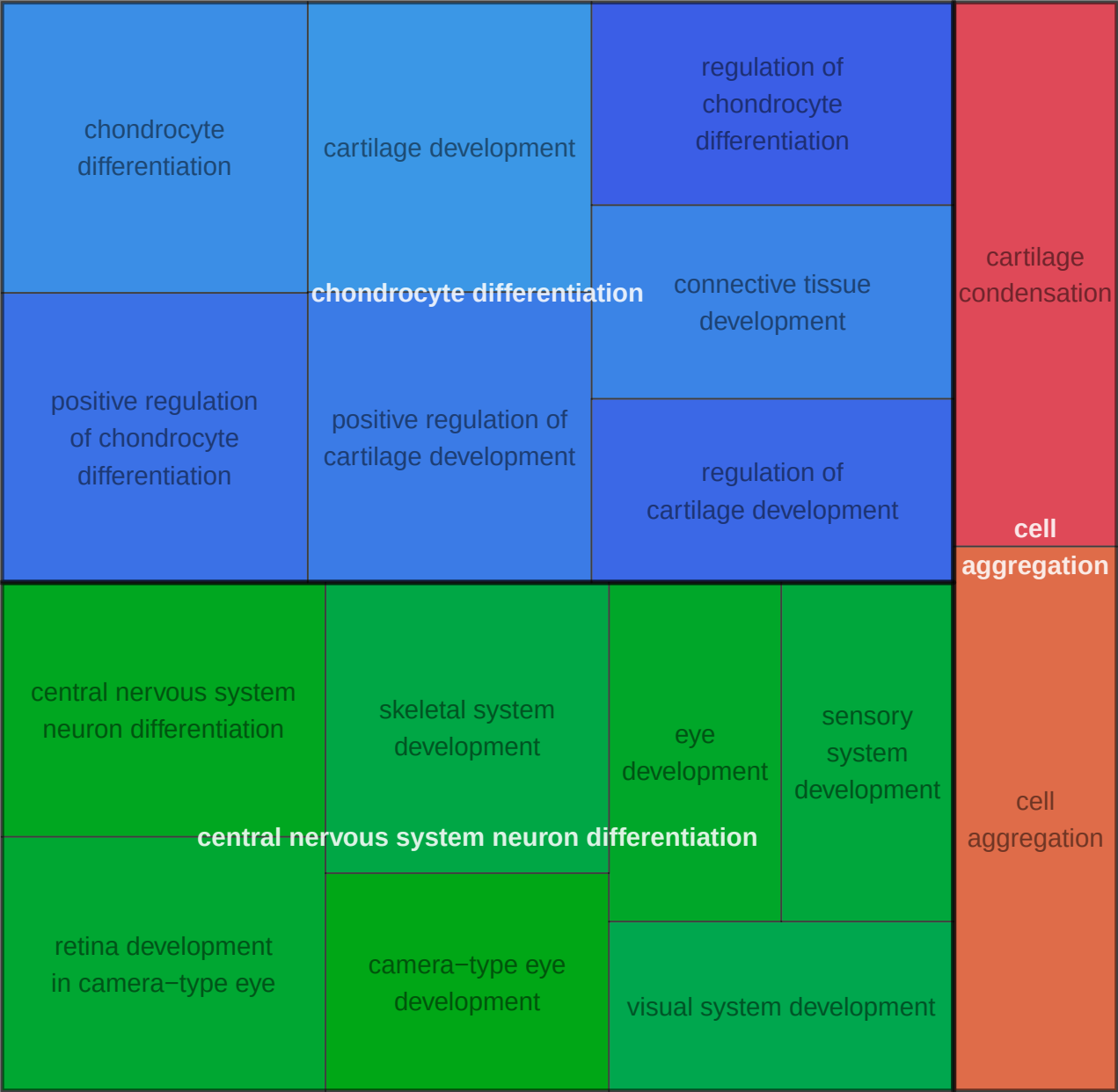
apical junction complex

cell-cell junction

**HFE-transferrin
receptor
complex**

cerebellar mossy fiber

Excit-L5-2_up_M BP



Inhib-Sst_down_ALL CC

cytosolic large ribosomal subunit

large ribosomal subunit

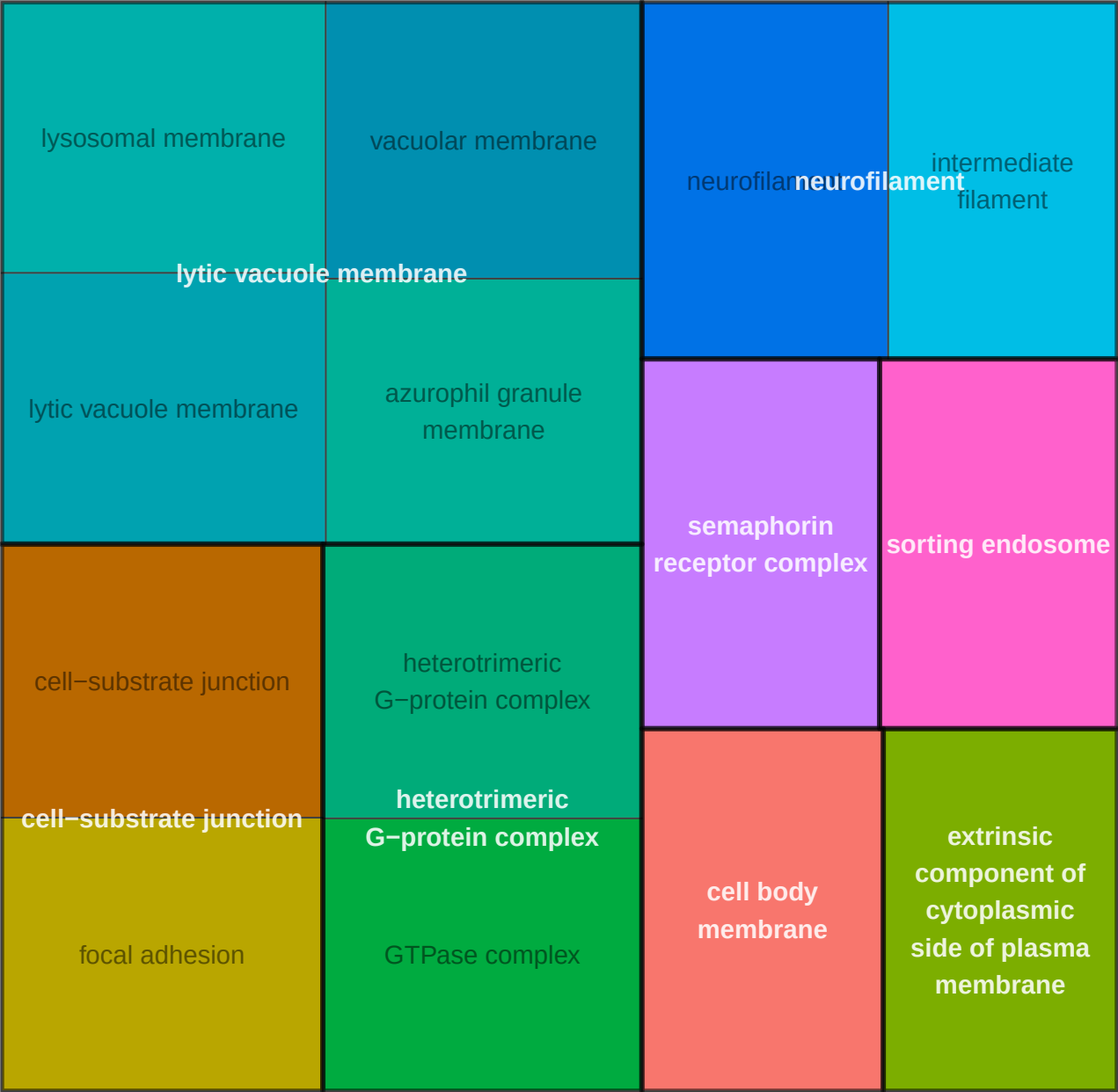
cytosolic ribosome

ribosomal subunit

cytosolic ribosome

ribosome

Macro_down_ALL CC



Macro_down_ALL BP



Macro_down_ALL MF



Macro_down_M CC

cell-substrate junction

semaphorin receptor complex

cell-substrate junction

focal adhesion

sorting endosome

neurofilament

cell body membrane

Macro_down_M BP



Macro_down_M MF



Macro_up_ALL CC

melanosome

pigment granule

pigment granule

Macro_up_ALL MF



Macro_up_M MF



Micro-4_up_ALL CC



Micro-4_up_ALL BP

modulation of chemical
synaptic transmission

regulation of trans-synaptic signaling

regulation of trans-synaptic signaling

Oligo-3_down_ALL CC



Oligo-3_down_ALL BP

axon ensheathment

myelination

ensheathment of neurons

myelination

Oligo-3_down_M CC

myelin sheath **myelin sheath**

compact myelin

neuromuscular junction

neuromuscular junction

cell-cell junction

main axon **main axon**

node of Ranvier

cell cortex

Oligo-3_down_M BP

axon ensheathment

myelination

axon
ensheathment
in central
nervous system

axon ensheathment

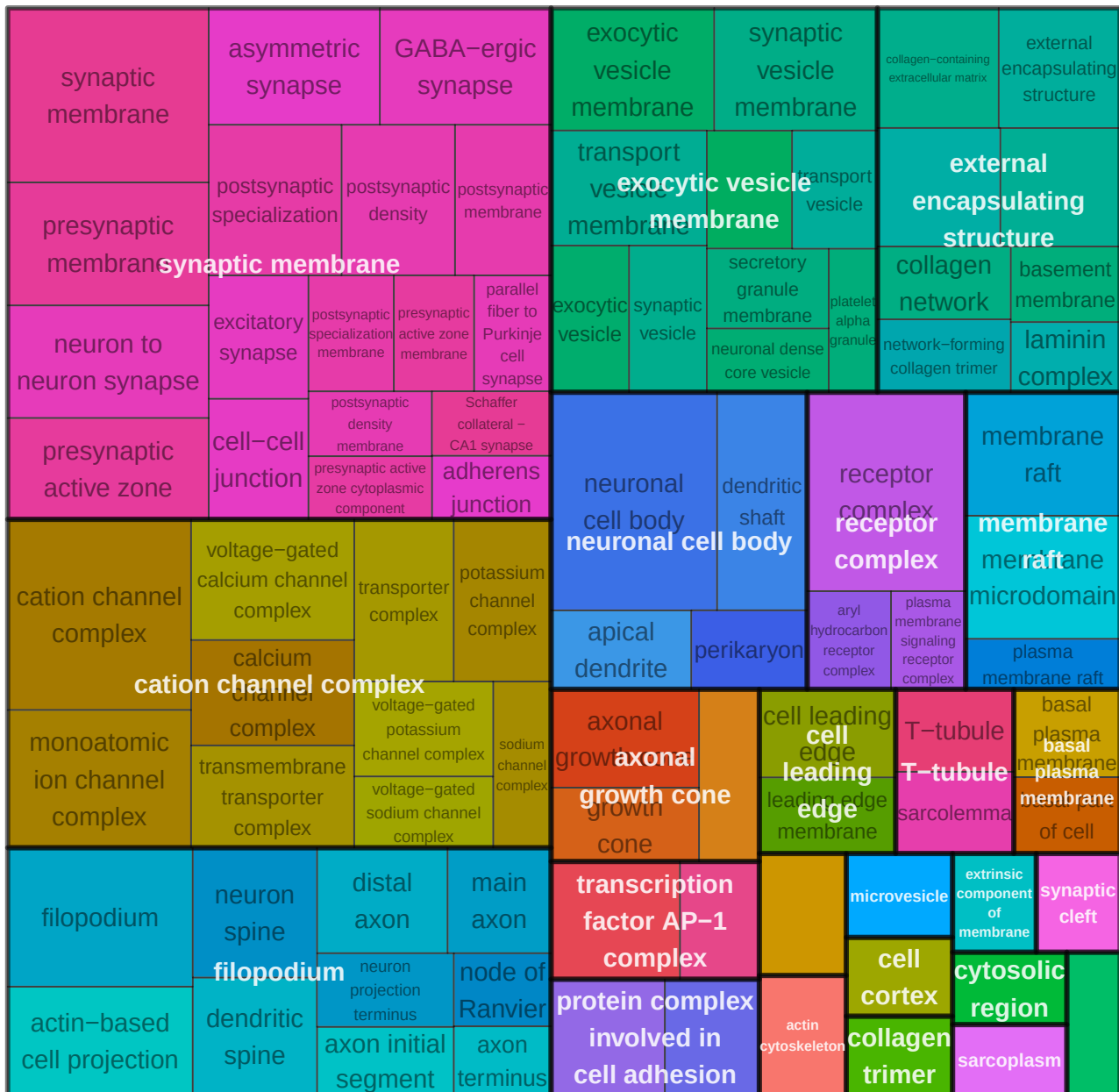
central nervous
system myelination

ensheathment of neurons

oligodendrocyte
development

myelin assembly

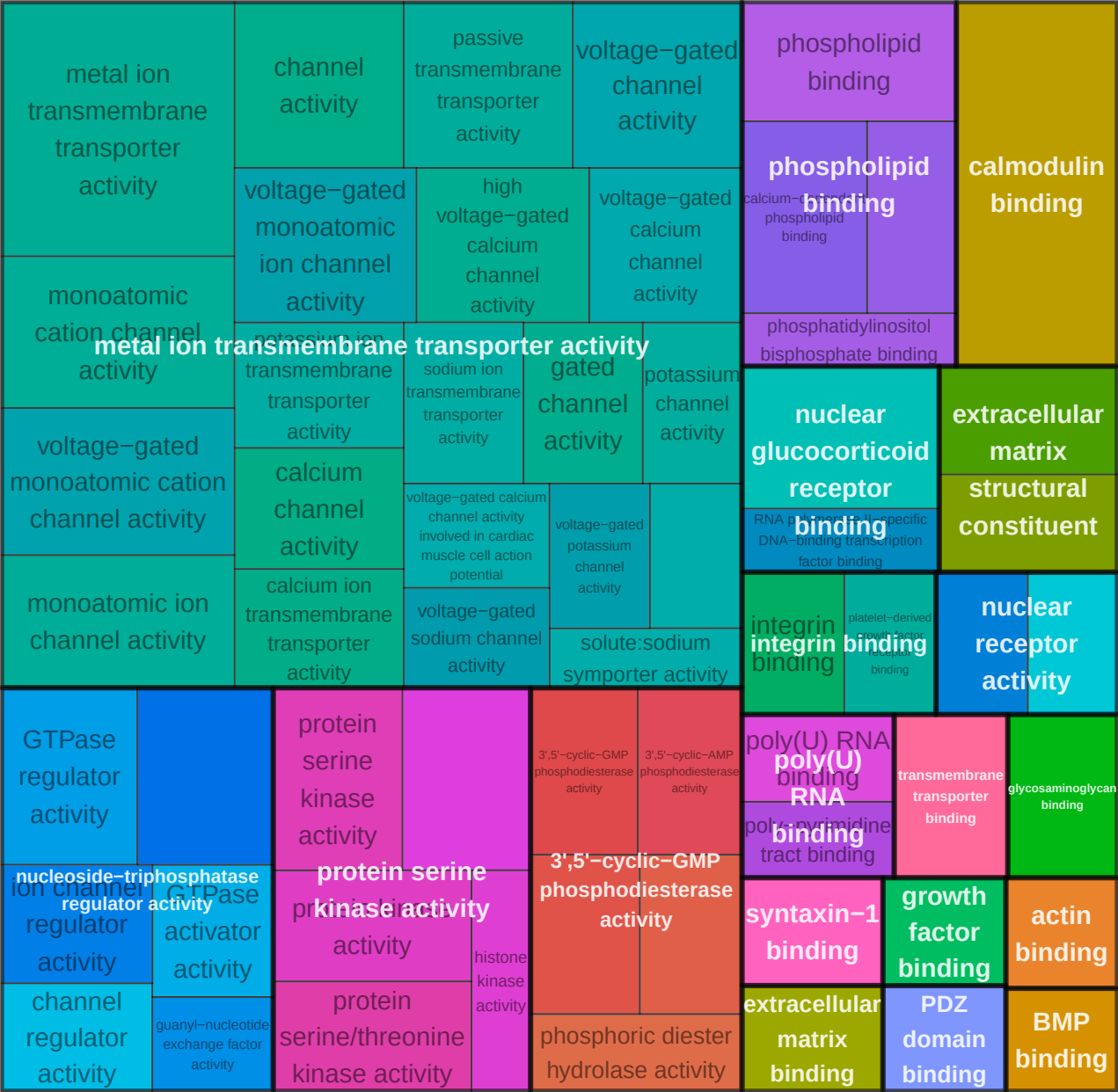
Oligo-3_up_ALL CC



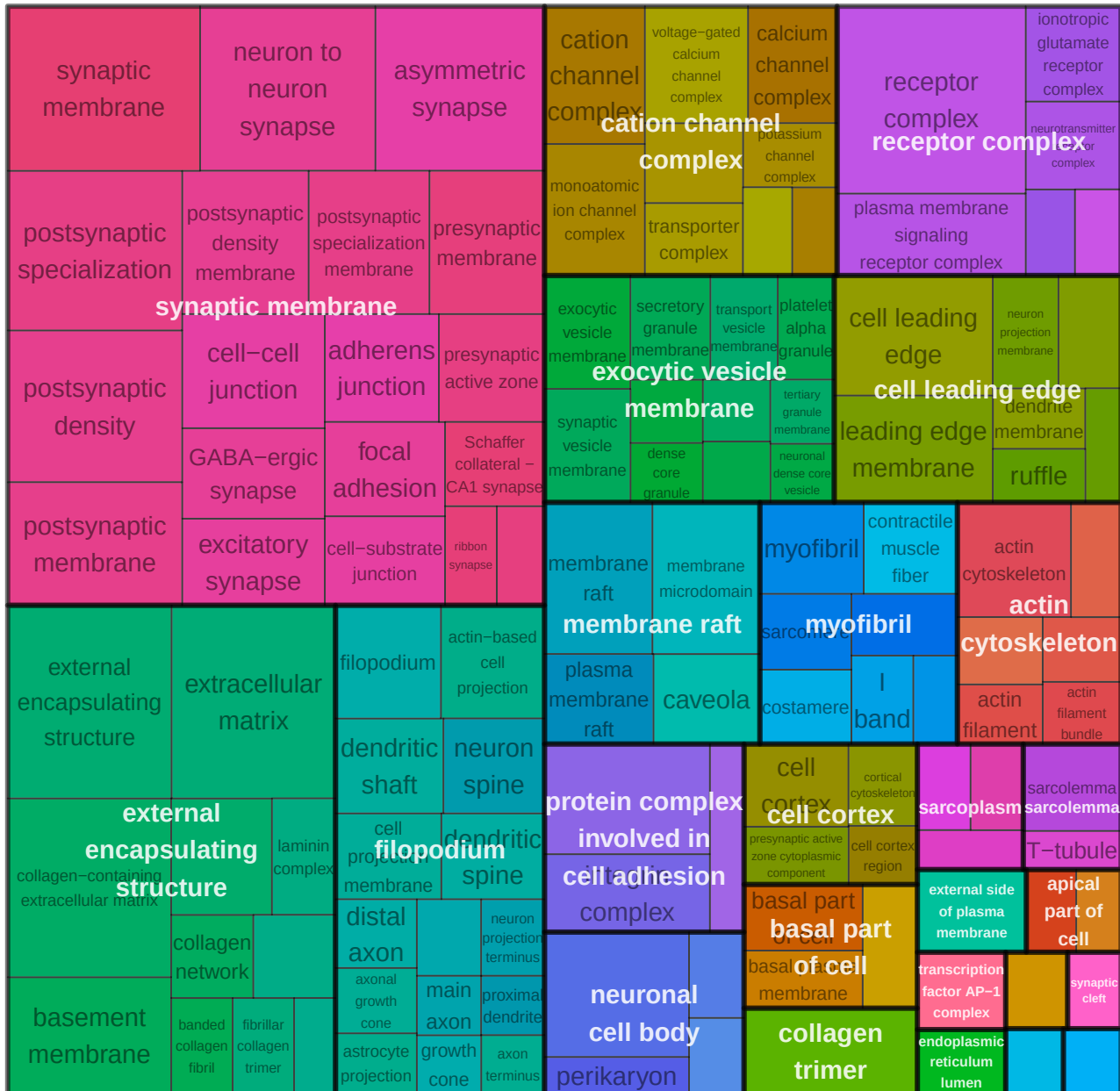
Oligo-3_up_ALL BP



Oligo-3_up_ALL MF



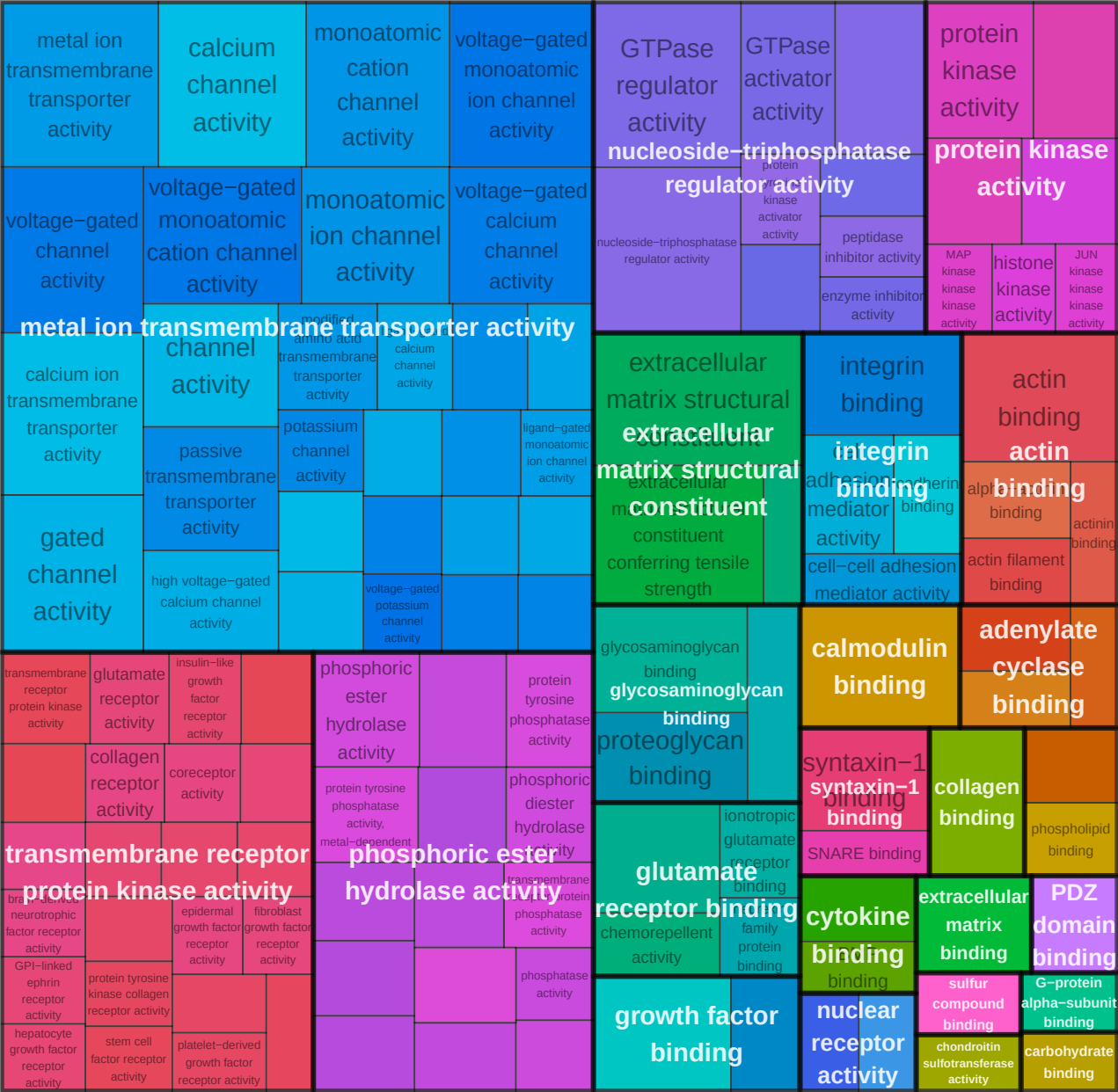
Oligo-3_up_M CC



Oligo-3_up_M BP



Oligo-3_up_M MF



Oligo-5_down_ALL CC

collagen type IX trimer

FACIT collagen trimer

FACIT collagen trimer

recycling endosome

Oligo-5_down_ALL MF



Oligo-5_down_M CC

collagen type IX trimer collagen type IX trimer FACIT collagen trimer

Oligo-5_up_ALL CC

membrane microdomain

membrane microdomain

membrane raft

immunological synapse

Oligo-5_up_ALL BP



Oligo-5_up_ALL MF

endopeptidase regulator activity

enzyme inhibitor activity

enzyme inhibitor activity

neurexin family
protein binding

```

graph TD
    A[membrane microdomain] --> B[membrane microdomain]
    A --> C[membrane raft]
    B --> D[cell-cell junction]
    B --> E[cell-substrate junction]
    C --> F[immunological synapse]
    C --> G[receptor complex]
    F --> H[phosphatidylinositol 3-kinase complex, class I]
    F --> I[phosphatidylinositol class I]
    G --> J[phosphatidylinositol class IA]
    G --> K[lamellipodium membrane]
  
```

membrane microdomain

membrane microdomain

membrane raft

cell-cell junction

cell-substrate junction

immunological synapse

receptor complex

phosphatidylinositol 3-kinase complex, class I

phosphatidylinositol class I

phosphatidylinositol class IA

lamellipodium membrane

membrane microdomain

membrane microdomain

membrane raft

cell-cell junctions

cell-substrate
junction

	1980	1985	1990	1995	2000	2005	2010	2015	2020
Population	76.5	80.5	84.5	88.5	92.5	96.5	100.5	104.5	108.5
GDP (constant prices)	10.5	15.5	20.5	25.5	30.5	35.5	40.5	45.5	50.5
Per capita GDP	137.1	192.5	242.6	289.2	329.8	367.8	402.9	435.0	465.0
Life expectancy at birth	65.0	68.0	71.0	74.0	77.0	80.0	83.0	86.0	89.0
Infant mortality rate	120.0	100.0	80.0	60.0	40.0	20.0	10.0	5.0	2.0
Fertility rate	5.0	4.5	4.0	3.5	3.0	2.5	2.0	1.5	1.0
Urban population (%)	30.0	35.0	40.0	45.0	50.0	55.0	60.0	65.0	70.0
Rural population (%)	70.0	65.0	60.0	55.0	50.0	45.0	40.0	35.0	30.0
Employment in agriculture (%)	40.0	35.0	30.0	25.0	20.0	15.0	10.0	5.0	2.0
Unemployment rate (%)	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0
Government expenditure as % of GDP	10.0	12.0	14.0	16.0	18.0	20.0	22.0	24.0	26.0
Private investment as % of GDP	15.0	18.0	21.0	24.0	27.0	30.0	33.0	36.0	39.0
Foreign direct investment as % of GDP	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Trade openness (% of GDP)	20.0	25.0	30.0	35.0	40.0	45.0	50.0	55.0	60.0
Corruption index	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
Human Development Index	0.500	0.550	0.600	0.650	0.700	0.750	0.800	0.850	0.900

immunological synapse

phosphatidylinositol
3-OH kinase complex, class I
class IA

receptor complex

lamellipodium membrane

Oligo-5_up_M BP



Oligo-5_up_M MF

protein kinase activity

transcription coactivator binding

transmembrane receptor protein
protein kinase activity
tyrosine kinase activity

transcription coactivator binding

transcription coregulator binding

vascular endothelial growth factor receptor activity

OPC-5_up_F CC

anaphase-promoting complex

anaphase-promoting complex

nuclear ubiquitin ligase complex

cullin-RING ubiquitin
ligase complex

OPC-5_up_F BP



[illegible]

sarcomere

Z disc

platelet dense
tubular network
membrane

myofibril

I band

complex of
collagen
trimers

cell cortex

presynaptic
active zone
cytoplasmic
component

contractile actin
filament bundle

actin filament
bundle

actomyosin

receptor complex

receptor complex

**collagen
trimer**

plasma membrane
signaling
receptor complex

contractile actin filament bundle

stress fiber

muscle myosin
complex

myosin II
complex

cell-cell junction

**sarcoplasmic
reticulum**

Vasc-Endo_up_M BP



Vasc-Endo_up_M MF



Vasc-VLMC_down_ALL MF



Vasc-VLMC_up_M CC

external encapsulating structure

extracellular matrix

external encapsulating structure

desmosome

collagen-containing extracellular matrix